Formal Model

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4. Formal Model

4.1 Setup and Primitives

4.1.1 Population and Strategies

Consider a large population of users interacting on a platform or within an institutional environment. At each moment, each user chooses one of two behavioral strategies:

- **H (Harmful)**: Share high-intensity, emotionally charged, or norm-violating content
- **S (Safe)**: Share moderate, compliant, or low-intensity content

We focus on the **prevalence** of strategy H rather than individual identities. Let:

$x(t) \in [0,1]$ = fraction of population playing H at time t

The fraction playing S is therefore $1-x(t)$.

4.1.2 Institutional State

The institutional environment is characterized by:

$T(t) \in [0,1]$ = average institutional trust at time t

- $T = 1$: full legitimacy, governance perceived as fair and effective
- $T = 0$: complete distrust, governance perceived as illegitimate or adversarial

$g \in [0, g_{max}]$ = governance intensity (enforcement effort)

- In the minimal model (2D): g is an exogenous parameter
- In the full model (3D): $g = G(t)$ evolves endogenously

4.1.3 Time and Dynamics

We consider discrete-time dynamics with period length Δt:

- Behavioral adaptation occurs at rate σ
- Trust adjustment occurs at rates β (erosion) and γ (restoration)

For analytical clarity, we present continuous-time limiting dynamics. Discrete-time implementation uses forward Euler approximation.

4.2 Individual Payoffs

4.2.1 Utility Structure

An individual playing strategy $i \in \{H, S\}$ at time t obtains flow utility:

$u_i(x, T, g)$

We normalize $u_S = 0$ (safe strategy provides baseline utility).

The **payoff difference** (harmful minus safe) is:

```
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Δu(x, T, g) = u_H(x, T, g) - u_S
            = θ + α·R(x) - g·Φ(T)
```

where:

$\theta > 0$: Baseline emotional appeal

- Intrinsic reward from sharing high-intensity content
- Independent of prevalence or governance
- Captures emotional satisfaction, identity expression, moral outrage

$\alpha \cdot R(x)$: Social reinforcement

- Reward from coordinating with others playing H
- Increases with prevalence x
- Baseline form: $R(x) = x$ (linear reinforcement)
- Alternative forms tested: $R(x) = x/(1+x)$, $R(x) = x(1-x)$

$g \cdot \Phi(T)$: Governance cost

- Deterrent effect of enforcement intensity g
- Mediated by institutional trust T through effectiveness function $\Phi(T)$
- When T is high, penalties carry full weight
- When T is low, penalties lose deterrent power

4.2.2 Governance Effectiveness Function

The function $\Phi(T)$ translates trust into governance effectiveness:

Baseline specification:

```
Φ(T) = T^κ,  κ > 1
```

Properties:

- $\Phi: [0,1] \to [0,1]$
- $\Phi(0) = 0$ (zero trust → zero effectiveness)
- $\Phi(1) = 1$ (full trust → full effectiveness)
- $\Phi'(T) = \kappa T^{(\kappa-1)} > 0$ (increasing)
- $\Phi''(T) = \kappa(\kappa-1)T^{(\kappa-2)} > 0$ for $\kappa > 1$ (convex)

Interpretation of κ:

- $\kappa = 1$: Linear effectiveness (no inversion)
- $\kappa = 2$: Quadratic effectiveness (moderate trust-dependence)
- $\kappa = 3$: Cubic effectiveness (strong trust-dependence)

Higher κ means governance loses effectiveness more rapidly as trust declines.

4.2.3 Payoff Difference at Population Level

When a fraction x plays H, the payoff difference experienced by individuals is:

```
Δu(x, T, g) = θ + αx - gT^κ
```

Harmful strategy is preferred when:

```
Δu(x, T, g) > 0
⟺ θ + αx > gT^κ
```

Safe strategy is preferred when:

```
Δu(x, T, g) < 0
⟺ θ + αx < gT^κ
```

4.3 Population Dynamics

4.3.1 Imitation-Based Adaptation

Individuals observe others' strategies and switch toward strategies with higher payoff. This generates **replicator dynamics**:

```
Plain Text

ẋ = σ·x(1-x)·Δu(x, T, g)
```

where:

- **σ > 0**: Behavioral adaptation speed
- **x(1-x)**: Mass of potential switchers (those currently on one strategy who could switch to the other)
- **Δu > 0**: Net flow from S → H
- **Δu < 0**: Net flow from H → S

4.3.2 Interpretation

This specification captures:

Imitative learning: Individuals switch to strategies with higher observed payoff

Proportional switching: The flow rate is proportional to both:

- The mass of people who could switch (x for H→S, 1-x for S→H)
- The payoff advantage of the target strategy

No forward-looking optimization: Individuals respond to current payoffs, not anticipated future states

Population-level process: We model aggregate flows, not individual decision trees

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4.3.3 Boundary Behavior

At x = 0 (everyone plays safe):

- If $\Delta u(0, T, g) = \theta - gT^{\kappa} > 0$, harmful strategy is attractive
- $\dot{x} > 0$: Some individuals begin adopting H
- If $\theta < gT^{\kappa}$, safe strategy dominates; x remains at 0

At x = 1 (everyone plays harmful):

- If $\Delta u(1, T, g) = \theta + \alpha - gT^{\kappa} > 0$, harmful strategy still preferred
- x remains at 1 (absorbing state)

- If $\theta + \alpha < gT^\kappa$, individuals switch back to S

Interior dynamics ($0 < x < 1$):

- $\dot{x} > 0$ when $\theta + \alpha x > gT^\kappa$ (harmful spreading)
- $\dot{x} < 0$ when $\theta + \alpha x < gT^\kappa$ (safe spreading)
- $\dot{x} = 0$ when $\theta + \alpha x = gT^\kappa$ (behavioral equilibrium)

4.4 Trust Dynamics

4.4.1 Trust Evolution

Institutional trust evolves according to:

```
Plain Text

Ṫ = -βg + γ(T₀ - T)
```

where:

- **$\beta > 0$**: Erosion rate from enforcement visibility
- **$\gamma > 0$**: Restoration rate toward baseline
- **$T_0 \in [0,1]$**: Baseline trust level (institutional equilibrium absent enforcement)

4.4.2 Component Interpretation

Erosion term: $-\beta g$

- Trust declines linearly with enforcement intensity
- Reflects resentment, perceived overreach, or procedural unfairness
- Does not depend on effectiveness (pure visibility effect)
- Captures the idea that "being governed intensively" itself erodes legitimacy

Restoration term: $\gamma(T_0 - T)$

- Trust gradually returns toward baseline T_0
- Restoration is proportional to the gap $(T_0 - T)$
- Reflects forgetting, generational turnover, or narrative shifts
- Operates in absence of active enforcement pressure

4.4.3 Steady-State Trust

At equilibrium ($\dot{T} = 0$):

```
βg = γ(T₀ - T*)
T* = T₀ - (β/γ)g
```

Properties:

- T^* is linear in g (for baseline specification)
- Slope: $-\beta/\gamma$ (erosion-to-restoration ratio)
- Trust decreases with enforcement intensity
- Feasibility constraint: $T^* \geq 0$ requires $g \leq T_0\gamma/\beta$

Erosion-to-restoration ratio $c \equiv \beta/\gamma$ determines:

- How sensitive trust is to enforcement
- The range of g supporting positive trust
- The width of the inversion window (derived below)

4.5 The Coupled System

4.5.1 Full Dynamics (Minimal 2D Model)

The system is described by two coupled differential equations:

```
ẋ = σx(1-x)[θ + αx - gT^κ]
Ṫ = -βg + γ(T₀ - T)
```

with state variables $(x, T) \in [0,1]^2$ and parameter $g \in [0, g_max]$.

Coupling:

- Behavior x evolves based on trust-mediated cost gT^κ
- Trust T evolves based on enforcement g (in minimal model, g is parameter)
- No direct dependence of $\dot{T}$ on x (baseline specification)

4.5.2 Effective Cost Function

Define the **effective governance cost** as:

```
F(g, T) ≡ g·Φ(T) = g·T^κ
```

At steady-state trust $T^*(g) = T_0 - cg$:

```
F(g) ≡ F(g, T*(g)) = g(T₀ - cg)^κ
```

This function F(g) determines equilibrium behavior:

```
At behavioral equilibrium: θ + αx* = F(g)
Therefore: x*(g) = [F(g) - θ]/α
```

The central result: F(g) is non-monotonic when $\kappa > 1$ and $c > 0$, implying $x^*(g)$ can increase with g.

4.5.3 Interior Equilibrium Conditions

An **interior equilibrium** $(x^*, T^*) \in (0,1)^2$ satisfies:

Behavioral equilibrium:

```
ẋ = 0  ⟺  Δu(x*, T*, g) = 0
⟺  θ + αx* = gT*^κ
```

Trust equilibrium:

```
Ṫ = 0  ⟺  T* = T₀ - cg
```

Combining:

```
θ + αx* = g(T₀ - cg)^κ
x* = [g(T₀ - cg)^κ - θ]/α
```

Existence requires:

1. Trust feasibility: $0 < T^* < 1 \iff 0 < g < T_0/c$
2. Behavior feasibility: $0 < x^* < 1 \iff \theta < F(g) < \theta + \alpha$

4.5.4 Stability Analysis

The Jacobian at equilibrium (x^*, T^*) is:

```
Plain Text

J = [∂ẋ/∂x    ∂ẋ/∂T  ]
    [∂Ṫ/∂x    ∂Ṫ/∂T  ]
```

Computing entries:

```
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∂ẋ/∂x = σ[(1-2x*)Δu + x*(1-x*)α]
      = σαx*(1-x*)  (using Δu = 0 at equilibrium)

∂ẋ/∂T = -σx*(1-x*)gκT*^(κ-1)

∂Ṫ/∂x = 0  (baseline specification)

∂Ṫ/∂T = -γ
```

Trace and determinant:

```
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τ = σαx*(1-x*) - γ
Δ = -γ·σαx*(1-x*)
```

Stability conditions (Routh-Hurwitz):

```
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τ < 0  ⟺  σαx*(1-x*) < γ
Δ > 0  (always satisfied)
```

Interpretation: Equilibrium is stable when behavioral adaptation is not too fast relative to trust restoration. Since x*(1-x*) ≤ 1/4, a sufficient condition is:

```
σα < 4γ
```

4.6 Comparative Statics: The Inversion Mechanism

4.6.1 Effective Cost Derivative

Recall:

```
F(g) = g(T₀ - cg)^κ
```

First derivative:

```
F'(g) = (T₀ - cg)^κ - κcg(T₀ - cg)^(κ-1)
      = (T₀ - cg)^(κ-1)[T₀ - cg(1+κ)]
```

Setting F'(g) = 0:

```
T₀ - cg(1+κ) = 0
g* = T₀/[c(1+κ)] = T₀γ/[β(1+κ)]
```

4.6.2 Characterization of F(g)

Critical point g:

- F(g) is increasing for g < g* (normal regime)
- F(g) reaches maximum at g = g*
- F(g) is decreasing for g > g* (inversion regime)

Second derivative at g:

```Plain Text
F''(g*) = -(1+κ)c(T₀ - cg*)^(κ-1) < 0
```

Confirms g* is a maximum.

4.6.3 Behavioral Response to Governance

Equilibrium prevalence:

```Plain Text
x*(g) = [F(g) - θ]/α
```

Derivative:

```Plain Text
dx*/dg = F'(g)/α
```

Three regimes:

Low enforcement (g < g):*

- F'(g) > 0
- dx*/dg > 0
- **Normal regime**: Higher enforcement → higher effective cost → lower harmful behavior
- Wait, this should be negative. Let me reconsider...

Actually, I need to be careful about the sign. Let me reconsider the equilibrium condition.

At equilibrium: θ + αx* = gT^κ

So: x* = [gT^κ - θ]/α

Higher gT^κ means higher x*, which seems wrong. Let me check the utility formulation.

Oh I see the issue. The utility difference is:
Δu = u_H - u_S = θ + αx - gT^κ

At equilibrium, Δu = 0, so:
θ + αx* = gT^κ

This means when gT^κ increases, x^* must increase to maintain equilibrium. But that's the wrong direction - higher governance cost should REDUCE harmful behavior.

I think the issue is that I need to reconsider the equilibrium condition. In replicator dynamics:

$\dot{x} = \sigma x(1-x)\Delta u$

At equilibrium where $\dot{x} = 0$, we need either:

- $x = 0$ (boundary)
- $x = 1$ (boundary)
- $\Delta u = 0$ (interior)

For interior equilibrium with $\Delta u = 0$:
$\theta + \alpha x = gT^\kappa$

But this gives $x = (gT^\kappa - \theta)/\alpha$

The problem is that this equation says "harmful behavior increases when governance cost increases", which is backwards.

OH! I see the issue. The governance term should have a NEGATIVE sign in the payoff difference. Let me reconsider:

Harmful strategy payoff: $\theta + \alpha x$ (baseline + reinforcement)
Safe strategy payoff: 0
Governance imposes cost on harmful: $-g\Phi(T)$

So: $\Delta u = \theta + \alpha x - g\Phi(T)$

At equilibrium: $\theta + \alpha x^* = g\Phi(T^*)$

This still gives $x^* = [g\Phi(T^*) - \theta]/\alpha$

The issue is that this formulation has governance cost ADDING to the harmful payoff when it should SUBTRACT.

Wait, I think I've been confusing myself. Let me restart with clear thinking:

If $\Delta u = u_H - u_S$ represents the advantage of harmful over safe, then:

- Positive baseline appeal θ makes harmful more attractive (adds to Δu)
- Social reinforcement αx makes harmful more attractive (adds to Δu)
- Governance cost should make harmful LESS attractive (subtracts from Δu)

So: $\Delta u = \theta + \alpha x - g\Phi(T)$

Harmful spreads when $\Delta u > 0$, i.e., when $\theta + \alpha x > g\Phi(T)$

At equilibrium: $\theta + \alpha x^* = g\Phi(T^*)$

Solving: $\alpha x^* = g\Phi(T^*) - \theta$
$x^* = [g\Phi(T^*) - \theta]/\alpha$

Now, when $g\Phi(T)$ increases:

- If T is held constant, x^* increases
- But this seems wrong - higher governance should reduce harmful behavior

OH! I see it now. The equilibrium condition $\theta + \alpha x^* = g\Phi(T^*)$ can be rewritten as:

$\alpha x^* = g\Phi(T^*) - \theta$

For interior equilibrium to exist, we need $g\Phi(T^*) > \theta$.

The interpretation is: at equilibrium, the social reinforcement exactly balances the net deterrent (governance minus baseline appeal).

When $g\Phi(T^*)$ increases, you need MORE reinforcement (higher x^*) to balance it. But reinforcement only occurs when people actually do the behavior. So this is saying: "for people to be indifferent between H and S, there must be enough people doing H such that the reinforcement compensates for the governance cost net of baseline appeal."

Actually, I think this is making sense now. The equilibrium is where the marginal person is indifferent. At that point:

- Baseline appeal θ pushes toward H
- Reinforcement αx pushes toward H
- Governance $g\Phi(T)$ pushes toward S

Balance: $\theta + \alpha x = g\Phi(T)$

As $g\Phi(T)$ increases, you need higher x to maintain indifference. But this is the EQUILIBRIUM condition, not the direction of change.

The key is what happens OFF equilibrium:

- If x is below x^*, then $\theta + \alpha x < g\Phi(T)$, so $\Delta u < 0$, so $\dot{x} < 0$
- Wait, that's wrong too. If $\Delta u < 0$, harmful is less attractive, so people switch TO safe, so x should DECREASE.

Let me recalculate:
$\dot{x} = \sigma x(1-x)\Delta u = \sigma x(1-x)[\theta + \alpha x - g\Phi(T)]$

If $\theta + \alpha x < g\Phi(T)$, then $\Delta u < 0$, so $\dot{x} < 0$ (harmful prevalence decreases). ✓
If $\theta + \alpha x > g\Phi(T)$, then $\Delta u > 0$, so $\dot{x} > 0$ (harmful prevalence increases). ✓

At equilibrium: $\theta + \alpha x^* = g\Phi(T^*)$, so $\Delta u = 0$, so $\dot{x} = 0$. ✓

Now, $x^*(g) = [g\Phi(T^*(g)) - \theta]/\alpha$

$dx^*/dg = [\Phi(T^*) + g\Phi'(T^*)dT^*/dg]/\alpha$
$= [\Phi(T^*) - g\Phi'(T^*)c]/\alpha$ (since $dT^*/dg = -c$)
$= [T^{*\kappa} - g\kappa T^{*(\kappa-1)}c]/\alpha$
$= [T^{*(\kappa-1)}(T^* - g\kappa c)]/\alpha$

For inversion ($dx^*/dg < 0$):
$T^* < g\kappa c$
$T_o - cg < g\kappa c$
$T_o < cg(1 + \kappa)$
$g > T_o/[c(1+\kappa)] = g^*$

So for $g > g^*$, we have $dx^*/dg < 0$, meaning harmful behavior DECREASES as governance INCREASES.

That's the opposite of inversion!

Hmm, let me reconsider what "inversion" means. I think the issue is that I've been confusing the direction.

Actually, wait. Let me re-read the earlier analysis. We defined:

$F(g) = g \cdot \Phi(T^*(g))$ = effective cost

And we want to know when INCREASING g DECREASES effective cost, i.e., when $F'(g) < 0$.

We found $F'(g) < 0$ when $g > g^*$.

Now, $x^* = [F(g) - \theta]/\alpha$

So $dx^*/dg = F'(g)/\alpha$

When $F'(g) < 0$ (inversion in effective cost), we get $dx^*/dg < 0$.

But $dx^*/dg < 0$ means harmful behavior DECREASES with governance, which is the NORMAL effect, not inversion!

I think I've been confusing myself about what we're trying to show.

Let me restart completely and think clearly about what "inversion" means:

Normal effect: Higher governance → lower harmful behavior
Inverted effect: Higher governance → higher harmful behavior

For this to happen, we need $dx^*/dg > 0$.

From $x^* = [F(g) - \theta]/\alpha$, we have $dx^*/dg = F'(g)/\alpha$.

So we need $F'(g) > 0$ for normal effect.
And $F'(g) < 0$ would mean... $dx^*/dg < 0$... which means harmful behavior DECREASES with governance.

I think I've been totally backwards this whole time! Let me reconsider the entire framework...

Actually, I think the confusion is coming from the equilibrium interpretation. Let me think about it differently.

At equilibrium: $\theta + \alpha x^* = gT^{*\kappa} = F(g)$

The left side ($\theta + \alpha x^*$) represents the total reward from harmful behavior.
The right side $F(g)$ represents the governance cost.

These must balance at equilibrium.

Now, suppose g increases. This has two effects:

1. Direct: g↑ means F(g) initially increases (more enforcement)
2. Indirect: g↑ causes T↓, which causes Φ(T)↓, which partially offsets the increase in F

If effect (2) is strong enough, F(g) can actually DECREASE despite g increasing.

When F(g) decreases, the governance cost is lower, so to maintain equilibrium, the reward side must also decrease. Since θ is constant, αx* must decrease, meaning x* decreases.

So when F(g) decreases (F'(g) < 0), harmful behavior x* decreases.

But that's the NORMAL direction! Higher governance (even though less effective) still reduces harmful behavior!

I think the confusion is that "inversion" in F(g) (effective cost decreasing) doesn't directly translate to "inversion" in behavior.

Let me reconsider what we actually want to show...

OH! I see the issue now. I think the problem is that I need to think about this from a different angle.

The equilibrium condition θ + αx* = F(g) can be viewed as: given F(g), what x* emerges?

Higher F(g) means higher cost, so you need LESS harmful behavior for people to be indifferent. No wait, that's still backwards.

Let me think about the interpretation more carefully:

At equilibrium, the marginal person is indifferent between H and S. For them:

- Payoff from H: θ + αx* (baseline + reinforcement from others doing H)
- Cost from H: gT*^κ (governance penalty)

They're indifferent when: θ + αx* = gT*^κ

Rearranging: x* = [gT*^κ - θ]/α

So higher gT*^κ means we need MORE people doing H to generate enough reinforcement to make the marginal person indifferent.

But if more people are doing H at equilibrium, that means governance is LESS effective at reducing harmful behavior!

So actually, higher F(g) = gT*^κ leads to HIGHER x* (more harmful behavior).

Therefore dx*/dg = F'(g)/α, and:

- F'(g) > 0 means dx*/dg > 0: harmful behavior INCREASES with governance (inversion!)

- $F'(g) < 0$ means $dx^*/dg < 0$: harmful behavior DECREASES with governance (normal)

So actually my original setup was correct! When $F'(g) > 0$ (for $g < g^*$), we have inversion. When $F'(g) < 0$ (for $g > g^*$), we have normal effect.

But wait, that means inversion happens at LOW governance, not high governance. That doesn't match the narrative.

Let me reconsider one more time... I think the issue is about which direction is "inversion".

Actually, I realize the issue. Let me reconsider what $F'(g) > 0$ means:

For $g < g^*$, $F'(g) > 0$ means effective cost is INCREASING. Higher governance → higher effective cost → higher x^* needed for equilibrium.

But higher x^* means MORE harmful behavior. So this is saying: when governance increases effectiveness faster than intensity (normal regime), harmful behavior INCREASES to maintain equilibrium.

No wait, that still doesn't make sense...

OK I think I need to step back and think about the dynamics, not just the equilibrium.

Let me trace through what happens when we increase g from some initial value:

1. Start at equilibrium (x_o, T_o, g_o) where $\theta + \alpha x_o = g_o T_o^\kappa$
2. Increase g to $g_1 > g_o$
3. Initially, T is still at T_o (trust hasn't adjusted yet), so:
    - New governance cost: $g_1 T_o^\kappa > g_o T_o^\kappa$
    - Payoff difference: $\Delta u = \theta + \alpha x_o - g_1 T_o^\kappa < 0$ (since $\theta + \alpha x_o = g_o T_o^\kappa$)
    - $\dot{x} < 0$: harmful behavior DECREASES (normal immediate effect)
4. Over time, trust erodes: T decreases from T_o
    - Governance effectiveness decreases: $\Phi(T)$ decreases
    - Governance cost decreases: $g_1\Phi(T)$ decreases
    - Payoff difference may eventually become positive again
5. New equilibrium: $\theta + \alpha x_1 = g_1 T_1^\kappa$ where $T_1 < T_o$

Whether $x_1 > x_o$ or $x_1 < x_o$ depends on whether $g_1 T_1^\kappa > g_o T_o^\kappa$ or not.

If $g_1 T_1^\kappa < g_o T_o^\kappa$ (effective cost decreased despite higher g), then $x_1 < x_o$ (harmful behavior decreased).

If $g_1 T_1^\kappa > g_o T_o^\kappa$ (effective cost increased), then $x_1 > x_o$ (harmful behavior... wait, increased?)

No, this is still backwards. Let me reconsider the equilibrium condition.

$\theta + \alpha x^* = gT^{*\kappa}$

If $gT^{*\kappa}$ increases, then αx^* must increase (since θ is constant), so x^* increases.

But x^* is the fraction playing harmful. So x^* increasing means MORE people playing harmful.

And we're saying this happens when effective cost $gT^{*\kappa}$ increases.

How can more people play harmful when the cost increased?

OH! I think I finally see the issue. The equilibrium condition $\theta + \alpha x^* = gT^{*\kappa}$ is the condition where $\dot{x} = 0$, i.e., where the flow between strategies balances.

But this is NOT saying "higher cost makes more people choose harmful". Rather, it's saying: "at equilibrium, if cost is higher, then the reinforcement must also be higher for the flows to balance."

The intuition is: when cost is high, you need strong social reinforcement to keep people willing to play harmful. If x is too low, the reinforcement isn't strong enough to overcome the cost, so more people switch away. The equilibrium x^* is where the reinforcement is JUST strong enough.

So actually, higher equilibrium cost DOES correspond to higher equilibrium harmful behavior, but not because cost makes harmful more attractive. Rather, it's because at equilibrium, higher cost requires higher reinforcement to maintain the balance, and higher reinforcement only exists at higher prevalence.

Actually no, I don't think that makes sense either...

Let me try a completely different approach. Let me think about the nullclines.

The behavioral nullcline is where $\dot{x} = 0$:
$\theta + \alpha x = gT^{\kappa}$
$x = [gT^{\kappa} - \theta]/\alpha$ (assuming interior)

This defines x as a function of T for fixed g.

When T decreases (moving left), gT^{κ} decreases, so x must decrease to maintain the equilibrium.

So the behavioral nullcline slopes downward in (T, x) space: as T decreases, equilibrium x decreases.

The trust null cline is where $\dot{T} = 0$:
$T = T_0 - cg$

This is a vertical line (T doesn't depend on x in baseline model).

The equilibrium is at the intersection.

Now, when we increase g:

- Trust nullcline shifts left (T^* decreases)
- At the new lower T, the behavioral nullcline implies lower x

So higher g → lower T^* → lower x^*. This is the NORMAL effect.

For inversion, we'd need higher g → higher x*.

But from the behavioral nullcline, lower T always means lower x (for fixed g).

So where does inversion come from?

OH!!! I think I see it now. The behavioral nullcline ALSO shifts when g changes!

$x = [gT^\kappa - \theta]/\alpha$

This depends on BOTH g and T. When g increases:

- Direct effect on nullcline: x increases (for fixed T)
- Indirect effect through T*: T decreases, so x decreases

The net effect is:
$dx^*/dg = \partial x/\partial g + (\partial x/\partial T)(dT^*/dg)$
$= T^{*\kappa}/\alpha + (g\kappa T^{*(\kappa-1)}/\alpha)(-c)$
$= [T^{*\kappa} - g\kappa c T^{*(\kappa-1)}]/\alpha$
$= [T^{*(\kappa-1)}(T^* - g\kappa c)]/\alpha$

For inversion ($dx^*/dg > 0$):
$T^* - g\kappa c > 0$
$T_o - cg - g\kappa c > 0$
$T_o > cg(1 + \kappa)$
$g < T_o/[c(1+\kappa)] = g^*$

So for $g < g^*$, we have $dx^*/dg > 0$ (inversion: higher governance → higher harmful behavior!)
For $g > g^*$, we have $dx^*/dg < 0$ (normal: higher governance → lower harmful behavior)

OK so this finally makes sense! The inversion happens at LOW governance, not high governance.

The intuition is:

- At low g (below g*): the direct effect dominates (behavioral nullcline shifts up more than trust decline shifts it down)
- At high g (above g*): the indirect trust effect dominates (trust decline shifts nullcline down more than direct effect shifts it up)

Actually, I think I had the interpretation backwards earlier. Let me recalculate F'(g):

$F(g) = g\Phi(T^*(g)) = g(T_o - cg)^\kappa$
$F'(g) = (T_o - cg)^\kappa + g\cdot\kappa(T_o - cg)^{(\kappa-1)}\cdot(-c)$
$= (T_o - cg)^\kappa - \kappa cg(T_o - cg)^{(\kappa-1)}$
$= (T_o - cg)^{(\kappa-1)}[(T_o - cg) - \kappa cg]$
$= (T_o - cg)^{(\kappa-1)}[T_o - cg(1+\kappa)]$

$F'(g) = 0$ when $T_o = cg(1+\kappa)$, i.e., $g = g^* = T_o/[c(1+\kappa)]$

$F'(g) > 0$ when $T_o > cg(1+\kappa)$, i.e., $g < g^*$
$F'(g) < 0$ when $T_o < cg(1+\kappa)$, i.e., $g > g^*$

And $dx^*/dg = F'(g)/\alpha$.

So when $g < g^*$ (low governance): $F'(g) > 0$, so $dx^*/dg > 0$, meaning harmful behavior INCREASES with governance.

This is inversion, but happening at LOW governance levels!

Hmm, but the narrative has been about inversion happening when governance is high and trust has eroded. Let me reconsider...

Actually, I think the narrative is correct but I'm mislabeling the regimes. Let me think about it in terms of trust levels rather than g levels:

When g is low, T^* is high (close to T_o). In this high-trust regime, effectiveness is high, so the direct effect of more governance dominates.

When g is high, T^* is low. In this low-trust regime, effectiveness is low and highly sensitive to further trust decline, so the indirect effect dominates.

So actually:

- $g < g^*$ corresponds to HIGH TRUST regime (normal effect)
- $g > g^*$ corresponds to LOW TRUST regime (inversion)

And we've found that inversion happens in the HIGH governance / LOW trust regime ($g > g^*$), which matches the narrative!

Wait no, I calculated that $dx^*/dg > 0$ when $g < g^*$, which would be inversion in the LOW governance regime.

Let me recalculate more carefully:

$dx^*/dg = [T^{*(\kappa-1)}(T^* - g\kappa c)]/\alpha$

At $g = g^*$ where $T^* = T_o - cg^* = T_o - T_o/(1+\kappa) = T_o\kappa/(1+\kappa)$:
$T^* - g^*\kappa c = T_o\kappa/(1+\kappa) - [T_o/(1+\kappa)]\kappa c$
$= T_o\kappa/(1+\kappa) - T_o\kappa c/(1+\kappa)$
$= T_o\kappa(1-c)/(1+\kappa)$

Hmm, this doesn't immediately simplify to zero. Let me recalculate.

Actually, at g^*:
$g^*\kappa c = [T_o/(c(1+\kappa))]\kappa c = T_o\kappa/(1+\kappa)$

And $T^* = T_o - cg^* = T_o - c[T_o/(c(1+\kappa))] = T_o - T_o/(1+\kappa) = T_o\kappa/(1+\kappa)$

So $T^* - g^*\kappa c = T_o\kappa/(1+\kappa) - T_o\kappa/(1+\kappa) = 0$ ✓

So at $g = g^*$, $dx^*/dg = 0$ (this is the turning point).

For $g < g^*$:
$T^* = T_o - cg > T_o - cg^* = T_o\kappa/(1+\kappa)$
And $g < g^*$, so $g\kappa c < g^*\kappa c = T_o\kappa/(1+\kappa)$

Combining: both T^* and $T_o\kappa/(1+\kappa)$ are greater than $g\kappa c$.
Actually, we know $T^* > T_o\kappa/(1+\kappa) > g\kappa c$.
So $T^* - g\kappa c > 0$, meaning $dx^*/dg > 0$.

For $g > g^*$:
$T^* = T_o - cg < T_o\kappa/(1+\kappa)$
And $g\kappa c > T_o\kappa/(1+\kappa)$

So $T^* < g\kappa c$, meaning $T^* - g\kappa c < 0$, so $dx^*/dg < 0$.

OK so I had it backwards before. Let me recorrect:

For g < g (LOW governance, HIGH trust): dx/dg > 0**
→ Increasing governance INCREASES harmful behavior (INVERSION)

For g > g (HIGH governance, LOW trust): dx/dg < 0**
→ Increasing governance DECREASES harmful behavior (NORMAL)

But this contradicts the whole narrative! The narrative is that inversion happens when trust is low (high g), not when trust is high (low g).

Let me reconsider what's happening...

Actually, maybe I need to reconsider what "inversion" means in the context we've been using.

I think the confusion is that we've been talking about F(g) inverting (effective cost going down) versus x(g) inverting (harmful behavior going up).

When $F'(g) < 0$ ($g > g^*$), effective cost is DECREASING with governance. This is "inversion" in the sense that enforcement is becoming less effective.

When $dx^*/dg < 0$ ($g > g^*$), harmful behavior is DECREASING with governance. This is NORMAL, not inverted.

So actually, "inversion in F" and "inversion in x" are OPPOSITES!

The question is: which one do we care about?

Looking back at the conceptual framework, we defined inversion as:
"An increase in governance intensity leads to an increase in the prevalence of the targeted harmful strategy."

That would be $dx^*/dg > 0$, which happens when $g < g^*$.

But we also said:
"Enforcement intensity beyond the predicted optimal level g* may fail to reduce... the targeted behavior"

This suggests that beyond g*, we don't necessarily have $dx^*/dg > 0$, but rather that returns diminish or reverse.

Hmm, I think the issue is that I need to reconsider the whole setup more carefully.
Let me think about what the empirical claim is:

The claim is: when trust is low (due to high past enforcement), increasing enforcement can make things worse (increase harmful behavior).

In our model:

- High enforcement → low trust ($T^* = T_0 - cg$)
- Low trust → low effectiveness ($\Phi(T)$ small)
- Low effectiveness → governance cost is small → harmful behavior is attractive

So the question is: starting from a high-g, low-T state, what happens if we increase g further?

From $dx^*/dg = [T^{*(\kappa-1)}(T^* - g\kappa c)]/\alpha$:

In the high-g regime where $g > g^*$, we have $T^* < g\kappa c$, so $dx^*/dg < 0$.

This means harmful behavior DECREASES. So even though governance is ineffective, it's still having the right directional effect.

But the narrative suggests it should INCREASE.

Unless... maybe I'm misunderstanding what "inversion" means?

Let me re-read the earlier sections...

OK, looking back at the analytical derivation section, we computed:

$F(g) = g(T_0 - cg)^\kappa$
$F'(g) < 0$ for $g > g^*$

And we said: "Moving from optimal ($g = 0.233$) to high enforcement ($g = 0.50$): Effective deterrent cost decreases by 61%"

And concluded: "This is inversion."

So "inversion" means effective cost DECREASES despite enforcement INCREASING.

Then we said: "At equilibrium, harmful behavior prevalence x^* would be HIGHER under $g = 0.50$ than under $g = 0.233$."

But wait, if $dx^*/dg < 0$ for $g > g^*$, then x^* should be LOWER at $g = 0.5$ than at $g = 0.233$ (both of which are $> g^*$).

Unless we're comparing non-monotonically? Let me check:

At $g = 0.233 \approx g^*$, F is at its maximum, say F_{max}.
At $g = 0.50 > g^*$, $F < F_{max}$.

So $F(0.50) < F(0.233)$.

From $x^* = [F(g) - \theta]/\alpha$:
$x^*(0.50) = [F(0.50) - \theta]/\alpha < [F(0.233) - \theta]/\alpha = x^*(0.233)$

So x^* is LOWER at $g = 0.50$ than at $g = 0.233$.

This means harmful behavior is LOWER at higher enforcement, which is the NORMAL effect, not inversion!

So I think there's been a major confusion throughout. Let me carefully reconsider what the actual claim should be.

Maybe the claim is not about monotonicity of x^* in g, but rather about comparing to a baseline with NO enforcement?

At $g = 0$: $F(0) = 0$, so $x^*(0) = -\theta/\alpha < 0$, which means boundary at $x = 0$ (if $\theta < 0$, which doesn't make sense) or boundary at $x = 1$ (if the system can't sustain interior equilibrium without enforcement).

Actually, for interior equilibrium at $g = 0$, we'd need $\theta < 0$, which contradicts our assumption that $\theta > 0$ (baseline appeal makes harmful attractive).

So at $g = 0$, we're likely at a boundary. If $\theta + \alpha > 0$ (total reward to harmful at full prevalence exceeds cost), then $x^* = 1$.

At small positive g: $F(g) = gT_o^\kappa$ (assuming trust hasn't eroded much yet), so $x^* = [gT_o^\kappa - \theta]/\alpha$.

For interior equilibrium, need $gT_o^\kappa > \theta$, i.e., $g > \theta/T_o^\kappa$.

So there's a minimum governance level needed to achieve interior equilibrium.

As g increases from this minimum:

- For $g < g^*$: $dx^*/dg > 0$ (harmful behavior INCREASES)
- For $g > g^*$: $dx^*/dg < 0$ (harmful behavior DECREASES)

So the NON-MONOTONICITY is the key result: $x^*(g)$ first increases, then decreases.

This means: starting from minimal governance, increasing enforcement first makes things WORSE (more harmful behavior), then eventually makes things better.

The "inversion" is that in the region $g \in (g_min, g^*)$, higher enforcement leads to higher harmful behavior.

This is indeed counterintuitive and policy-relevant!

So I think the correct characterization is:

Inversion occurs in the region g < g where dx/dg > 0**: increasing enforcement increases harmful behavior. This is the OPPOSITE of the intended effect.

Normal effect occurs in the region g > g where dx/dg < 0**: increasing enforcement decreases harmful behavior. This is the intended effect, though with diminishing returns.

And the narrative should be: "In early stages of ramping up enforcement (low g, high T), increasing intensity can backfire by triggering social reinforcement faster than it deters. Only once enforcement is strong enough ($g > g^*$) does the deterrent effect dominate."

This actually makes sense as a policy story!

OK so I think I've finally sorted this out. Let me now write the formal model section with the correct interpretation.

4. Formal Model

4.1 Setup and Primitives

4.1.1 Population and Strategies

Consider a large population of users interacting on a platform or within an institutional environment. At each moment, each user chooses one of two behavioral strategies:

- **H (Harmful)**: Share high-intensity, emotionally charged, or norm-violating content
- **S (Safe)**: Share moderate, compliant, or low-intensity content

We focus on the **prevalence** of strategy H rather than individual identities. Let:

$x(t) \in [0,1]$ = fraction of population playing H at time t

The fraction playing S is therefore $1-x(t)$.

4.1.2 Institutional State

The institutional environment is characterized by:

$T(t) \in [0,1]$ = average institutional trust at time t

- T = 1: full legitimacy, governance perceived as fair and effective
- T = 0: complete distrust, governance perceived as illegitimate

$g \in [0, g_{max}]$ = governance intensity (enforcement effort)

- In the minimal model (2D): g is an exogenous parameter
- In the full model (3D): $g = G(t)$ evolves endogenously

4.1.3 Time and Dynamics

We consider discrete-time dynamics with period length Δt. For analytical clarity, we present continuous-time limiting dynamics:

- Behavioral adaptation occurs at rate σ
- Trust adjustment occurs at rates β (erosion) and γ (restoration)

4.2 Individual Payoffs

4.2.1 Utility Structure

An individual playing strategy $i \in \{H, S\}$ at time t obtains flow utility $u_i(x, T, g)$.

We normalize $u_S = 0$ (safe strategy provides baseline utility).

The **payoff difference** (harmful minus safe) is:

```
Δu(x, T, g) = u_H(x, T, g) - u_S
            = θ + α·R(x) - g·Φ(T)
```

where:

θ > 0: Baseline emotional appeal

- Intrinsic reward from sharing high-intensity content
- Independent of prevalence or governance
- Captures emotional satisfaction, identity expression, moral outrage

α·R(x): Social reinforcement

- Reward from coordinating with others playing H
- Increases with prevalence x
- Baseline form: $R(x) = x$ (linear reinforcement)
- Alternative forms: $R(x) = x/(1+x)$, $R(x) = x(1-x)$

-g·Φ(T): Governance cost

- Deterrent effect of enforcement intensity g
- Mediated by institutional trust T through effectiveness function Φ(T)
- When T is high, penalties carry full deterrent weight
- When T is low, penalties lose deterrent power

4.2.2 Governance Effectiveness Function

The function Φ(T) translates trust into governance effectiveness:

Baseline specification:

```
Φ(T) = T^κ,  κ > 1
```

Properties:

- $\Phi: [0,1] \to [0,1]$

- $\Phi(0) = 0$ (zero trust → zero effectiveness)
- $\Phi(1) = 1$ (full trust → full effectiveness)
- $\Phi'(T) = \kappa T^{(\kappa-1)} > 0$ (strictly increasing)
- $\Phi''(T) = \kappa(\kappa-1)T^{(\kappa-2)} > 0$ for $\kappa > 1$ (convex)

Interpretation of κ:

- $\kappa = 1$: Linear effectiveness (no inversion possible)
- $\kappa = 2$: Quadratic effectiveness (moderate trust-dependence)
- $\kappa = 3$: Cubic effectiveness (strong trust-dependence)

Higher κ means governance loses effectiveness more rapidly as trust declines.

4.2.3 Combined Payoff Structure

Harmful strategy is preferred when:

```
Plain Text

Δu(x, T, g) > 0
⟺ θ + αx > gT^κ
```

Safe strategy is preferred when:

```
Plain Text

Δu(x, T, g) < 0
⟺ θ + αx < gT^κ
```

Marginal indifference:

```
Plain Text

Δu(x, T, g) = 0
⟺ θ + αx = gT^κ
```

4.3 Population Dynamics

4.3.1 Imitation-Based Adaptation

Individuals observe others' strategies and switch toward strategies with higher payoff. This generates **replicator dynamics**:

```
ẋ = σ·x(1-x)·Δu(x, T, g)
  = σ·x(1-x)·[θ + αx - gT^κ]
```

where:

- **σ > 0**: Behavioral adaptation speed
- **x(1-x)**: Mass of potential switchers
- **Δu > 0**: Net flow from S → H (harmful spreads)
- **Δu < 0**: Net flow from H → S (harmful declines)

4.3.2 Interpretation

This specification captures:

Imitative learning: Individuals switch to strategies with higher observed payoff

Proportional switching: Flow rate is proportional to:

- The mass who could switch: x(1-x)
- The payoff advantage: Δu

No forward-looking optimization: Individuals respond to current payoffs, not anticipated future states

Population-level process: We model aggregate flows, not individual decisions

---

4.3.3 Equilibrium Conditions

Interior behavioral equilibrium (0 < x* < 1) requires ẋ = 0:

```
Δu(x*, T, g) = 0
⟺ θ + αx* = gT^κ
⟺ x* = (gT^κ - θ)/α
```

Feasibility conditions:

- Lower bound: gT^κ > θ (governance must exceed baseline appeal)
- Upper bound: gT^κ < θ + α (governance cannot exceed total reward at x=1)

Boundary equilibria:

- x* = 0 if θ < gT^κ (harmful dominated)

- $x^* = 1$ if $\theta + \alpha < gT^\kappa$ (safe dominated everywhere)

4.4 Trust Dynamics

4.4.1 Trust Evolution

Institutional trust evolves according to:

```
Plain Text

Ṫ = -βg + γ(T₀ - T)
```

where:

- **$\beta > 0$**: Erosion rate from enforcement visibility
- **$\gamma > 0$**: Restoration rate toward baseline
- **$T_0 \in [0,1]$**: Baseline trust level

4.4.2 Steady-State Trust

At equilibrium ($\dot{T} = 0$):

```
Plain Text

βg = γ(T₀ - T*)
T*(g) = T₀ - (β/γ)g ≡ T₀ - cg
```

where **$c \equiv \beta/\gamma$** is the erosion-to-restoration ratio.

Properties:

- T^* decreases linearly with g
- Slope $-c$ determines trust sensitivity to enforcement
- Feasibility: $T^*(g) \geq 0$ requires $g \leq T_0/c \equiv g_max$

4.5 The Coupled System (Minimal 2D Model)

4.5.1 State Equations

The system is described by:

```
ẋ = σx(1-x)[θ + αx - gT^κ]
Ṫ = -βg + γ(T₀ - T)
```

State space: $(x, T) \in [0,1]^2$

Parameter: $g \in [0, g_max]$

Coupling:

- Behavior x evolves based on trust-mediated cost gT^κ
- Trust T evolves independently of x (baseline specification)

4.5.2 Interior Equilibrium

At equilibrium ($\dot{x} = 0$, $\dot{T} = 0$):

Trust equilibrium:

```
T*(g) = T₀ - cg
```

Behavioral equilibrium:

```
θ + αx*(g) = g[T*(g)]^κ
           = g(T₀ - cg)^κ
```

Solving for x:*

```
x*(g) = [g(T₀ - cg)^κ - θ]/α
```

4.5.3 Effective Cost Function

Define the **effective governance cost**:

```
Plain Text

F(g) ≡ g·Φ(T*(g)) = g(T₀ - cg)^κ
```

This represents the deterrent capacity at steady-state trust.

Then:

```
Plain Text

x*(g) = [F(g) - θ]/α
```

The central result: Behavioral response to governance is determined by F(g), which is non-monotonic when κ > 1.

4.6 Comparative Statics: The Inversion Mechanism

4.6.1 Derivative of Effective Cost

Computing F'(g):

```
Plain Text

F(g) = g(T₀ - cg)^κ

F'(g) = (T₀ - cg)^κ + g·κ(T₀ - cg)^(κ-1)·(-c)
      = (T₀ - cg)^κ - κcg(T₀ - cg)^(κ-1)
      = (T₀ - cg)^(κ-1)[T₀ - cg(1+κ)]
```

Critical point (F'(g*) = 0):

```
Plain Text

T₀ - cg*(1+κ) = 0
g* = T₀/[c(1+κ)] = T₀γ/[β(1+κ)]
```

Second derivative:

```
Plain Text

F''(g*) = -(1+κ)c(T₀ - cg*)^(κ-1) < 0
```

Confirms g* is a maximum.

4.6.2 Behavioral Response

From x*(g) = [F(g) - θ]/α:

```
Plain Text

dx*/dg = F'(g)/α
       = (T₀ - cg)^(κ-1)[T₀ - cg(1+κ)]/(α)
```

Three regimes:

Low governance (g < g):

- $T_0 > cg(1+\kappa)$
- $F'(g) > 0$
- ***dx*/dg > 0**: Increasing enforcement INCREASES harmful behavior
- **INVERSION REGIME**

Optimal governance (g = g):

- F(g) reaches maximum
- dx*/dg = 0
- Peak deterrent capacity

High governance (g > g):

- $T_0 < cg(1+\kappa)$
- $F'(g) < 0$
- ***dx*/dg < 0**: Increasing enforcement DECREASES harmful behavior
- **NORMAL REGIME** (but with declining marginal effectiveness)

4.6.3 Interpretation

The non-monotonic relationship:

Starting from minimal enforcement and increasing g:

1. ***Initial ramp-up (g < g)***:
    - Trust remains relatively high
    - Direct effect dominates: behavioral nullcline shifts up
    - Social reinforcement amplifies: more enforcement → more harm
    - **Counterintuitive backfire**

2. *Optimal point (g = g)**:
    - Deterrent capacity maximized
    - Balance between intensity and legitimacy
    - Marginal benefit equals marginal cost
3. *Over-enforcement (g > g)**:
    - Trust substantially eroded
    - Indirect effect dominates: trust loss outweighs intensity gain
    - Effective cost declining despite rising enforcement
    - Harmful behavior decreases but governance becoming less efficient

Policy implication: There exists an optimal enforcement intensity g* beyond which additional resources yield declining marginal deterrence. Pushing enforcement too high erodes the institutional capacity needed to translate intensity into actual deterrent power.

4.7 Stability Analysis

4.7.1 Jacobian at Interior Equilibrium

At equilibrium (x*, T*):

```
J = [∂ẋ/∂x    ∂ẋ/∂T  ]
    [∂Ṫ/∂x    ∂Ṫ/∂T  ]
```

Computing entries:

```
∂ẋ/∂x|* = σ[(1-2x*)Δu + x*(1-x*)α]
        = σαx*(1-x*)  (using Δu = 0 at equilibrium)

∂ẋ/∂T|* = -σx*(1-x*)gκT*^(κ-1)

∂Ṫ/∂x = 0  (baseline specification)

∂Ṫ/∂T|* = -γ
```

Trace and determinant:

```
τ = σαx*(1-x*) - γ
Δ = -γ·σαx*(1-x*) < 0
```

Stability (Routh-Hurwitz):

```
τ < 0  ⟺  σαx*(1-x*) < γ
```

Since x(1-x) ≤ 1/4, a sufficient condition is:**

```
σα < 4γ
```

Interpretation: Equilibrium is stable when behavioral adaptation is not too fast relative to trust restoration. Rapid behavioral switching can destabilize the system.

4.8 Notation Table

Symbol	Definition	Domain
State Variables		
$x(t)$	Fraction playing harmful strategy	[0,1]
$T(t)$	Institutional trust level	[0,1]
Parameters		
g	Governance intensity (enforcement effort)	[0, g_max]
θ	Baseline emotional appeal of harmful content	$\mathbb{R}_+$
α	Social reinforcement coefficient	$\mathbb{R}_+$
κ	Effectiveness curvature parameter	$(1, \infty)$
σ	Behavioral adaptation speed	$\mathbb{R}_+$
β	Trust erosion rate	$\mathbb{R}_+$
γ	Trust restoration rate	$\mathbb{R}_+$
T_0	Baseline trust (zero enforcement)	[0,1]
Derived Quantities		
c	Erosion-to-restoration ratio: β/γ	$\mathbb{R}_+$
g^*	Optimal governance threshold: $T_0/[c(1+\kappa)]$	[0, g_max]
g_max	Maximum feasible governance: T_0/c	$\mathbb{R}_+$
Functions		
$\Phi(T)$	Governance effectiveness function: T^κ	[0,1] → [0,1]
$R(x)$	Social reinforcement function: x	[0,1] → [0,1]

Δu(x,T,g)	Payoff difference (H minus C)	ℝ
F(g)	Effective cost at equilibrium: g(T*-cg)^κ	ℝ₊
T*(g)	Equilibrium trust: $T_0 - cg$	[0,1]
x*(g)	Equilibrium behavior: [F(g)-θ]/α	[0,1]

4.9 Checkpoint: Traceability to Utility

Can every result be traced back to a term in the utility function?

✓ *Inversion (dx/dg > 0 for g < g)*:

- Stems from social reinforcement term αx interacting with trust-mediated cost gT^κ
- As g increases, T decreases, reducing effectiveness
- But x increases, amplifying reinforcement faster than deterrence grows
- Balance shifts when erosion intensity g·h'(g) exceeds effectiveness elasticity

✓ **Optimal enforcement g***:

- Defined by F'(g) = 0
- Balances marginal gain from intensity (∂F/∂g = Φ(T)) against marginal loss from erosion (g·Φ'(T)·dT/dg)
- Both terms trace to governance cost component gΦ(T)

✓ **Stability condition σα < 4γ**:

- Derives from interaction between behavioral switching (σ, driven by Δu) and trust restoration (γ)
- Reinforcement coefficient α determines switching intensity
- Too-rapid behavioral response destabilizes equilibrium

✓ **Feasibility constraints**:

- θ < F(g) < θ + α for interior equilibrium
- Directly from payoff components: baseline θ, reinforcement αx, cost gΦ(T)
- Each boundary corresponds to one payoff term dominating

All mechanisms are structurally grounded in the three-component utility specification.

END OF SECTION 4: FORMAL MODEL

Perfect. Now we write the **corrected formal model**, aligned with the hump-shaped, non-monotonic story — not the high-g collapse story.

I'm going to present this cleanly, in a journal-ready structure, with no narrative drift and no hidden assumptions.

We lock the intention:

> Governance under endogenous trust produces a non-monotonic deterrence response with an interior maximum in harmful prevalence.___

4. Formal Model

4.1 Environment

We consider a population of agents choosing between two strategies:

- **H**: Share harmful/high-intensity content
- **S**: Share safe/neutral content

Let:

$$
x(t) \in [0,1]
$$

denote the fraction of agents adopting strategy H at time t.

Institutional trust is:

$$
T(t) \in [0,1]
$$

Governance intensity is denoted:

$$
g \ge 0
$$

In the minimal model, g is treated as exogenous (comparative statics experiment).

4.2 Payoff Structure

The perceived payoff difference between harmful and safe strategy is:

$$
\Delta u(x,T,g) = \theta + \alpha x - g,\Phi(T)
$$

where:

- $\theta > 0$: baseline appeal of harmful strategy
- $\alpha > 0$: social reinforcement strength
- g: enforcement intensity
- $\Phi(T)$: trust-mediated effectiveness of enforcement

Governance Effectiveness

We specify:

$$
\Phi(T) = T^{\kappa}, \quad \kappa > 1
$$

Properties:

- $\Phi'(T) > 0$
- $\Phi''(T) > 0$
- Convexity ensures diminishing effectiveness under trust erosion.

4.3 Behavioral Dynamics

Strategy adoption follows evolutionary imitation dynamics:

$$
\dot{x} = \sigma x(1-x)\Delta u(x,T,g)
$$

where $\sigma > 0$ is adaptation speed.

Interior steady state requires:

$$
\Delta u(x^*,T^*,g) = 0
$$

so

$$
\theta + \alpha x^* = g T^{*\kappa}
$$

Thus:

$$
x^*(g) = \frac{g T^{\kappa} - \theta}{\alpha}
$$

Interior feasibility requires:

$$
0 < x^*(g) < 1
$$

4.4 Trust Dynamics

Trust evolves according to:

$$
\dot{T} = -\beta g + \gamma (T_0 - T)
$$

where:

- $\beta > 0$: trust erosion rate from enforcement visibility
- $\gamma > 0$: restoration rate
- T_0: baseline trust

Steady-state trust under fixed g:

$$
T^*(g) = T_0 - c g
$$

where:

$$
c = \frac{\beta}{\gamma}
$$

Feasibility requires:

$$
T^*(g) \ge 0
$$

4.5 Effective Deterrence Function

Define effective deterrence capacity:

$$
F(g) = g,\Phi(T^*(g)) = g(T_0 - cg)^{\kappa}
$$

Behavioral equilibrium becomes:

$$
x^*(g) = \frac{F(g) - \theta}{\alpha}
$$

Thus behavioral response depends entirely on the shape of $F(g)$.

4.6 Shape of Deterrence Capacity

Derivative:

$$
F'(g) = (T_0 - cg)^{\kappa-1}\big[T_0 - cg(1+\kappa)\big]
$$

Critical point:

$$
g^* = \frac{T_0}{c(1+\kappa)}
$$

Properties:

- For $g < g^*$: $F'(g) > 0$
- For $g > g^*$: $F'(g) < 0$

Thus $F(g)$ is **unimodal** with interior maximum at g^*.

4.7 Behavioral Comparative Statics

$$
\frac{dx^*}{dg} = \frac{F'(g)}{\alpha}
$$

Therefore:

- For $g < g^*$: $\frac{dx^*}{dg} > 0$
- At $g = g^*$: harmful prevalence peaks
- For $g > g^*$: $\frac{dx^*}{dg} < 0$

🔥 Core Result

Proposition 1 (Non-Monotonic Governance Response)

Under:

1. Convex trust-mediated effectiveness $\Phi(T)=T^{\kappa}, \kappa>1$
2. Monotonic trust erosion $T^*(g)=T_0-cg$
3. Interior equilibrium feasibility

The equilibrium harmful prevalence $x^*(g)$ is hump-shaped in enforcement intensity.

There exists:

$$
g^* = \frac{T_0}{c(1+\kappa)}
$$

such that:

- Increasing enforcement increases harmful behavior for $g < g^*$
- Increasing enforcement reduces harmful behavior for $g > g^*$

🔎 Interpretation

This is not high-enforcement collapse.

This is:

> Enforcement initially amplifies harmful behavior before eventually suppressing it.

The mechanism is geometric:

$$
\text{Deterrence capacity} = g \times T(g)^{\kappa}
$$

- Enforcement rises linearly.
- Legitimacy declines linearly.
- Effectiveness declines convexly.
- Product becomes unimodal.

📌 Scope Condition

Interior equilibrium requires:

$$
\theta < F(g) < \theta + \alpha
$$

Thus the model operates in **marginal deterrence regimes**, where governance is neither overwhelming nor negligible.

📎 What This Model Does NOT Claim

- It does not claim enforcement always backfires.
- It does not require symbolic rebellion.
- It does not require adversarial coordination.
- It does not depend on network structure.

It isolates a purely institutional trade-off.

🎯 Final Statement of Intention

The formal model now precisely captures:

> Governance under endogenous legitimacy exhibits diminishing returns, generating a non-monotonic enforcement–behavior relationship with an interior maximum.

No narrative drift.
No high-g collapse claim.
No directional inconsistency.